IMPORTANT **Definition of the Max-value Entropy Search policy**

By using this expected-reduction-in-entropy quantity as the acquisition score, we obtain the max-value entropy search (or MES) policy. The term "max-value" denotes the fact that we are using information theory to search for the max value, or the optimal value f*, of the objective function.

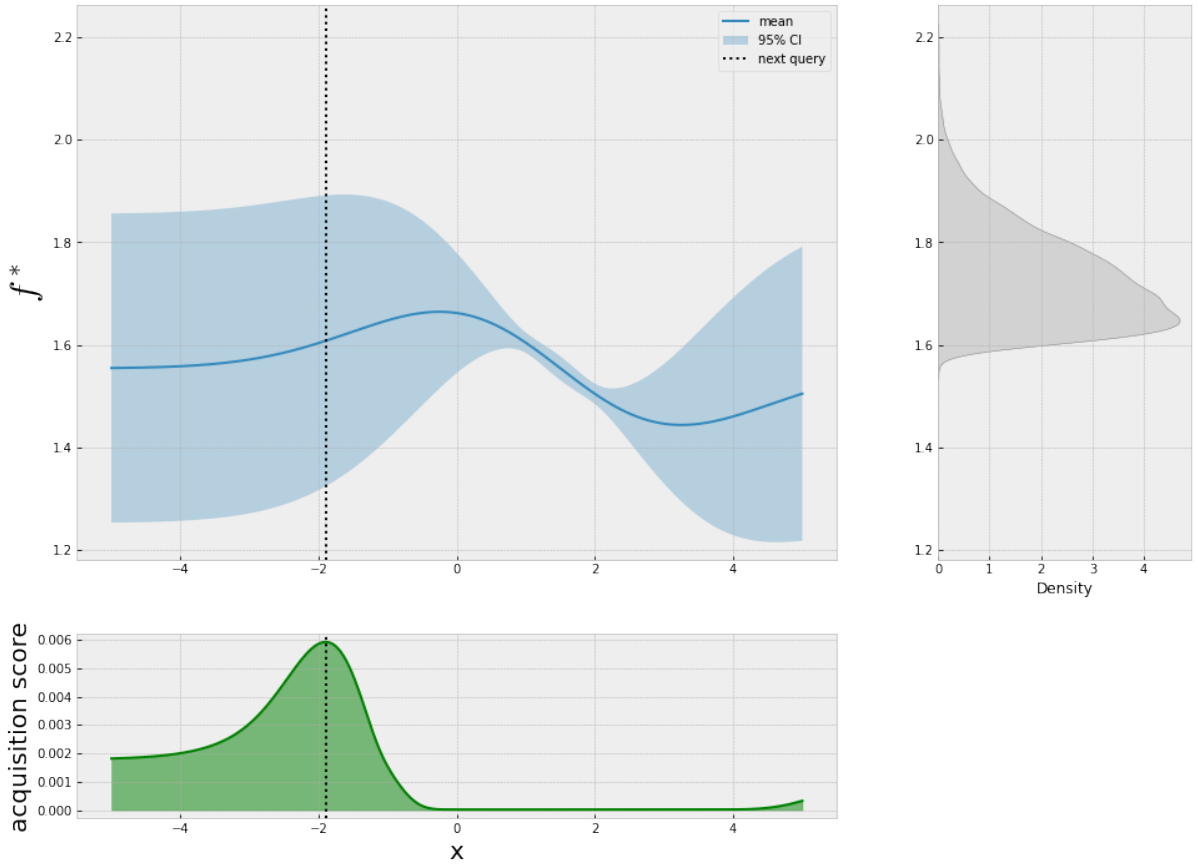


Figure 6.12 Gaussian process belief (top left), the distribution of the optimal value f* (right), and the approximate expected reduction in entropy, which is used as the acquisition function score (bottom). The distribution of the optimal value is always one-dimensional and therefore easier to work with than that of the optimizer.

Figure 6.12 shows the MES acquisition score in our running example in the bottom panel, where the point around -2 is where we should query next according to this information-theoretic criterion. The MES policy prefers this location because it has both a relative high mean and high CI, thus balancing between exploration and exploitation.

Interestingly, the acquisition landscape here looks somewhat similar to the distribution of the optimizer x^* itself, which is shown in Figure 6.11, where we see that the curve:

1. reaches its zenith somewhere in the middle,
2. achieves non-negligible value at the end points, and
3. bottoms out at zero around 2 where we know for certain that the location is not optimal.

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